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Increased body sway at 3.5-8 Hz in patients with phobic postural vertigo

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Abstract

Postural sway during upright stance was analyzed in 12 patients with phobic postural vertigo (PPV) and in 12 age-matched healthy volunteers. Recordings were made under different conditions (with the eyes open or closed): when standing on a foam rubber pad with the head upright, turned 30 degrees to the right or left, or during 1 Hz horizontal head oscillations. Sway analysis included calculation of sway path, of root mean square values of sway and of the power spectrum of sway in fore/aft and lateral directions. There was a significant increase in sway activity in the 3.53-8 Hz frequency band in patients with PPV. This did not impair objective postural stability. Increase in higher frequency sway activity may simply reflect a change in postural strategy rather than a sensorimotor dysfunction. The patients' conscious control of stance may augment co-activation of anti-gravity muscles, a strategy applied by normal subjects when performing demanding balancing tasks.

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